

# PhD Lecture in Behavioral Asset Pricing: Preferences & Beliefs (II)

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# Psychology of Beliefs

# Psychology of Beliefs: Representativeness

- Consider questions such as:
  - What is the probability that object A comes from class B?
  - What is the probability that event A was generated from process B?
- Kahneman and Tversky (1974) argue that people often answer by using the “representativeness heuristic”
  - Evaluate the probability by the extent to which A is representative of B
  - i.e. degree to which A reflects the essential characteristics of B
- This is helpful in general but can lead to serious biases

- **Base-rate neglect**
- Consider the following description:
  - Steve is very shy and withdrawn, invariably helpful, but with little interest in people or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail.
- People judge Steve to be more likely a librarian than a farmer
- But there are 20 times more farmers than librarians in the U.S.

$$\frac{p(\text{librarian}|\text{data})}{p(\text{lawyer}|\text{data})} = \frac{p(\text{data}|\text{librarian})}{p(\text{data}|\text{lawyer})} \frac{p(\text{librarian})}{p(\text{lawyer})}.$$

- **Sample size neglect, ctd.**
- When we do not know the model generating the data, we may infer it too quickly based on too little data
- This can lead to over-extrapolation of recent trends
- E.g., the hot hand fallacy
  - Seeing a basketball player make a few shots in a row leads to the belief that he is “hot”
  - But empirical evidence suggests that the hot hand is largely a myth (Gilovich, Vallone, and Tversky, 1985)

# Psychology of Beliefs: Overconfidence

- At least two distinct types:
- **Overprecision**
  - People are too confident in the accuracy of their forecasts
  - 98% confidence intervals include the true quantity around 60% of the time (Alpert and Raiffa, 1982)
- **Overplacement**
  - People think they are better than others
  - 90% of drivers think they are above-average drivers (Svenson, 1981)

# Overprecision: Survey Evidence

- Ben-David, Graham and Harvey (2013)
- CFO's expected stock market return distribution is too narrow
- Realized market returns are within the executives' 80% confidence intervals only 36% of the time

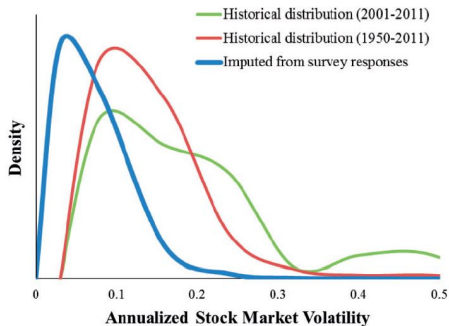


FIGURE IIA

One-Year Imputed Individual S&P 500 Volatility Versus Historical Distribution

The kernel density of the one-year S&P 500 return volatility for the period of 1950–2000 (or 2001–2011), and the kernel density of the imputed one-year S&P 500 individual volatility derived from CFOs' predictions.

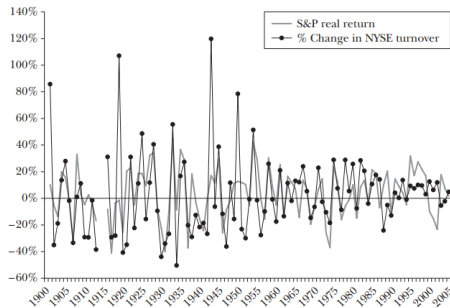
# Overconfidence: Excessive Trading

- **Excessive trading**

- Under efficient markets, no trading other than liquidity needs and portfolio rebalancing
- Speculative trading reflects differences of opinion and some amount of overconfidence
- Trading leads to lower returns
  - The more you trade, the lower your return (Odean, 1999)

Figure 3

Market-Wide Stock Returns and Changes in Turnover, 1901–2005



Hong and Stein (2007)

# Overconfidence: Excessive Trading

- **Boys will be boys**
- Barber and Odean (2001) show that men trade more frequently than women, but earn lower return
- Consistent with psychological studies that men are more overconfident than women (Lundeberg et al., 1994)

| Dependent variable | Mean monthly turnover (%) | Own-benchmark abnormal net return | Portfolio volatility |
|--------------------|---------------------------|-----------------------------------|----------------------|
| Intercept          | 6.269***<br>(4.24)        | -0.321***<br>(-11.47)             | 11.466***<br>(58.85) |
| Single             | 0.483***<br>(4.24)        | 0.002<br>(0.14)                   | 0.320***<br>(3.40)   |
| Woman              | -1.461***<br>(-12.76)     | 0.058***<br>(4.27)                | -0.689***<br>(-7.27) |
| Single × woman     | -0.733***<br>(-3.38)      | 0.027<br>(1.08)                   | -0.439***<br>(-2.45) |



# Overconfidence: Momentum

- Daniel, Hirshleifer, and Subrahmanyam (1998)
- Momentum is driven by investor overconfidence and self-attribution bias
  - Investors are overconfident in the private information they gather
  - Investors also suffer from self-attribution bias: they credit themselves for successes but blame external factors for failures
- Positive returns reinforce overconfidence, leading to more aggressive trading and price momentum
- Eventually, prices correct when information is revealed

# Overconfidence: Momentum

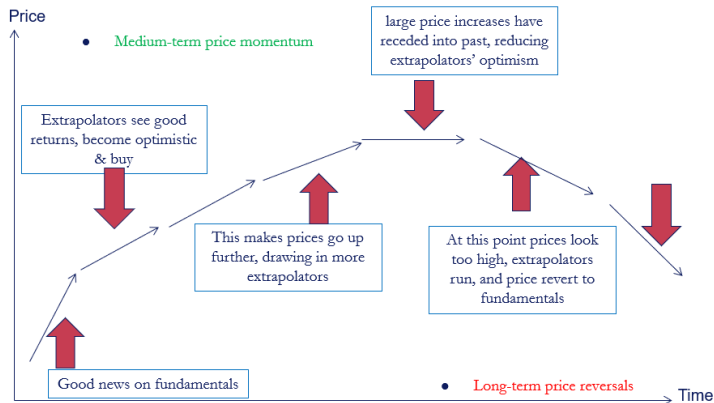
- Chui, Titman, and Wei (JF 2010) test the idea that momentum is driven by investor overconfidence and self-attribution bias
- They use an individualism index developed by Hofstede (2001) to proxy for investor overconfidence
- They find that momentum is stronger in countries with higher individualism scores

| Portfolio Formed Method             | Index on Individualism | Winner (W)      | Loser (L)         | W Minus L       |
|-------------------------------------|------------------------|-----------------|-------------------|-----------------|
| Panel A: Country-Average Portfolios |                        |                 |                   |                 |
| Country-average                     | Low                    | 1.628<br>(4.30) | 1.241<br>(2.96)   | 0.387<br>(2.80) |
|                                     | 2                      | 1.693<br>(5.22) | 1.004<br>(2.90)   | 0.689<br>(5.91) |
|                                     | High                   | 1.748<br>(6.14) | 0.707<br>(2.27)   | 1.041<br>(7.93) |
|                                     | High minus low         | 0.120<br>(0.39) | -0.534<br>(-1.53) | 0.654<br>(4.30) |
|                                     |                        |                 |                   |                 |

# Psychology of Beliefs: Over-extrapolation

- Estimate of the future value of a quantity is a positive function of the recent past values of that quantity
- Most important and widely-applied ideas in the behavioral finance
- The source of over-extrapolation: representativeness heuristic
- Application to finance:
  - Return extrapolation: predict future returns based on past returns
  - Fundamentals extrapolation: predict future earnings/cash flows based on past growth

# Over-extrapolation: Conceptual Framework

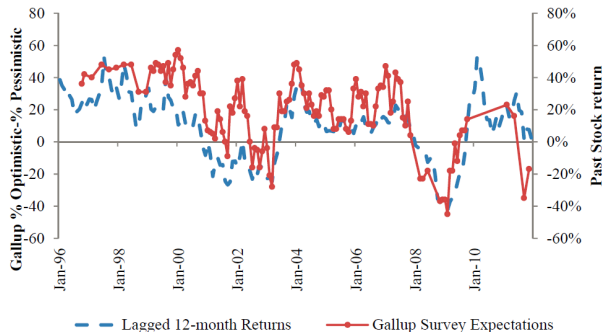


# Over-extrapolation: Survey Evidence

- **Excess volatility and predictability in aggregate stock market**
- Extrapolators buy assets whose prices have risen because they expect them to keep rising
- This leads to overvaluation and return predictability

The Role of Past Stock Market Returns in Explaining Survey Expectations

The dashed line denotes the 12-month rolling nominal return on the CRSP VW stock index. The solid line marked with circles denotes expectations from the Gallup survey (% optimistic - % pessimistic).



Greenwood and Shleifer (2014)

# Over-extrapolation: Survey Evidence

- Investors act on their over-extrapolative beliefs with flows into (out of) stock markets

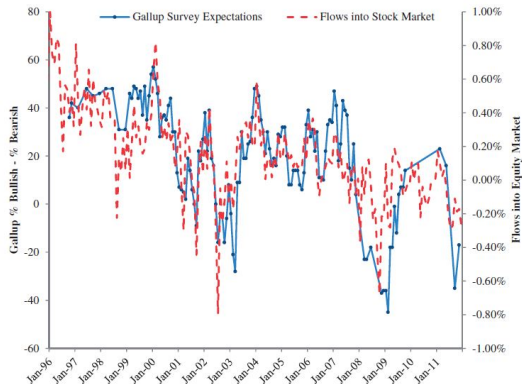


Figure 4

Comparing the Gallup survey with flows into equity mutual funds

The solid line denotes the percentage of investors who are bullish in the Gallup survey (left axis). The dashed line (right axis) is flows into mutual funds as a percentage of equity market capitalization, as reported by the Investment Company Institute.

Greenwood and Shleifer (2014)

# Over-extrapolation of Past Returns: Applications

- Using survey data on expectations of stock returns, Cassella and Gulen (RFS 2018) estimate the degree of extrapolative thinking in investors' beliefs (DOX) about market return
- DOX  $(1 - \lambda)$  measures the weight investors place on past returns

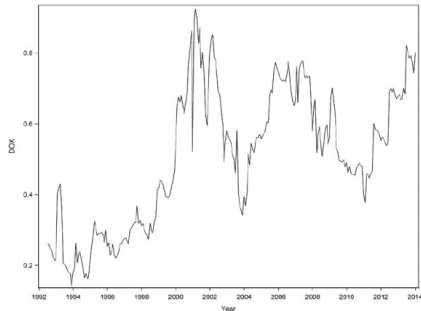


Figure 2

Time-varying DOX

The figure plots the recursively estimated DOX time series for the principal component (PC) of the II and AA surveys, in the sample period 1992:06–2013:12. (Details on estimation are in Section 2.)

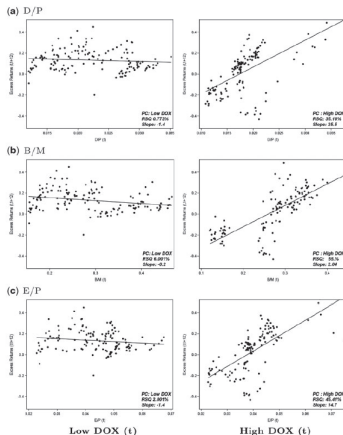
$$Exp_t = a + b \sum_{i=0}^{\infty} w_i R_{t-(i+1)\Delta t, t-i\Delta t},$$

$$w_i = \frac{\lambda^i}{\sum_{k=0}^{\infty} \lambda^k}, \quad 0 \leq \lambda < 1,$$

A lower  $\lambda$  implies that investors place higher weight on more recent observations, while earlier observations contribute less to an extrapolator's expectations

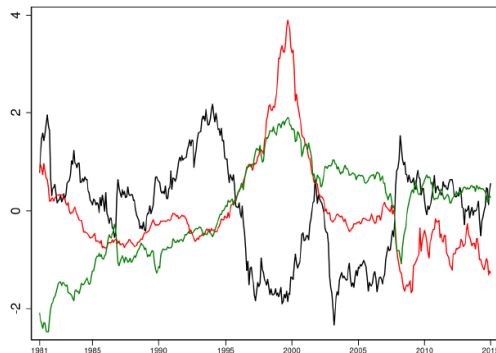
# Over-extrapolation of Past Returns: Applications

- They find the predictability of stock returns by price-scaled variable (e.g., D/P, B/M or E/P) is contingent on the DOX
- When DOX is high, investors more quickly “forget” the positive price changes that drove up valuations, making them less prone to extrapolate recent returns
- As a result, high valuations are more informative about low future returns when DOX is high



# Over-extrapolation of Past Fundamentals: Application

- Bordalo et al. (2023): overreaction of long-term earnings growth (LTG) expectation help explain time-series variation in P/D ratio and predict aggregate market return
- Good news cause investors to become overly optimistic about future earnings growth



This figure graphs LTG (red) and the price dividend ratio (green) together with realized stock returns in the coming five years (black)

Bordalo et al. (2023)

# Psychology of Beliefs: Conservatism

- Investors with extrapolative or overconfident beliefs adjust their views too much in response to new information and cause excessive movements in prices
- But some phenomena — momentum and PEAD — may instead require that investors adjust their views *too little*
- **Conservatism:** Individuals are too slow to update their beliefs in response to new information (Edwards, 1968)

# Psychology of Beliefs: Belief Perseverance and Confirmation Bias

- **Belief perseverance:**

- Once we have formed an opinion, we are often too slow to change it on receipt of new evidence
- We don't look for evidence that would falsify our beliefs
- We ignore evidence that goes against our beliefs

- **Confirmation bias:**

- We actively seek out evidence that supports our beliefs
- We interpret ambiguous evidence as supportive of our beliefs

# Psychology of Beliefs: Anchoring

- Kahneman and Tversky (1974): in forming numerical estimates of uncertain quantities, adjustments in assessments away from some initial value (called the “anchor”) are often insufficient
- E.g. estimates of the percentage of African countries in the UN are affected by a random number generated by spinning a wheel
- Anchoring effects persist even when the anchor is clearly irrelevant
- Applications in finance: IPO pricing, analyst forecasts, merger valuations

# The Effect of Feelings

- A recent strand of research investigates how feelings affect beliefs and preferences
  - Affective feelings (mood, emotion)
  - Cognitive feelings
- **Affective feelings**
  - A happier mood caused by an exogenous stimulus leads to more optimistic judgments and choices
  - A sadder mood leads to more pessimistic judgments
  - E.g. sunshine, sports outcomes, holidays

- **Sports**

- Edmans, Garcia, and Noli (2006): When the national soccer team loses a World Cup match, the national stock market falls the next day
- Disappointing sports news worsens the national mood, leading to lower stock returns
- The effect is stronger in countries where soccer is more important

- **Terrorist attacks**
- Wang and Yang (2020): Terrorist attacks lead to increased aggregate risk aversion, as proxied by flows into (out of) government bond funds (equity funds)
- These findings point to a mechanism by which feelings affect risk preferences

- Feeling of ease when accessing or processing information leads to more positive judgements than justified by content alone (Zajonc, 1968; Bornstein, 1989)
- E.g. people like familiar things more than justifiably
- **Applications:**
  - Home bias in portfolio choice
  - Local bias in stock ownership
  - Preference for stocks with pronounceable ticker symbols

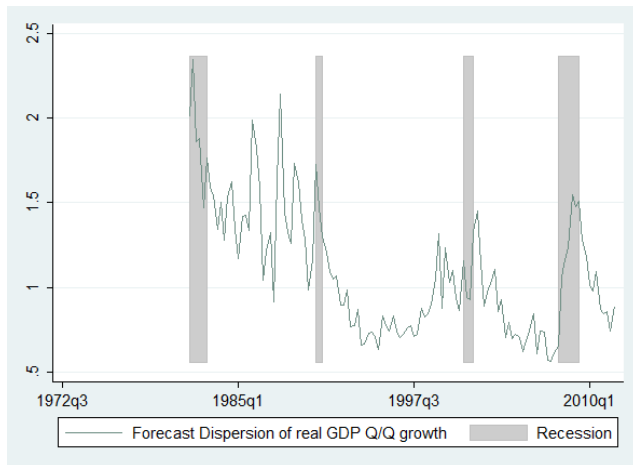
# Heterogeneous Beliefs

# Heterogeneous Beliefs: Overview

- Previously we look at the psychological foundations of systematically biased beliefs
  - Assuming representative investor
- In this part, we look at models with heterogeneous beliefs
  - Average belief is unbiased
  - But cross-sectional variation in beliefs can still affect prices

# Heterogeneous Beliefs, ctd.

- People regularly hold different beliefs about almost everything in life
  - from political views and sport events to economic outcomes and asset returns

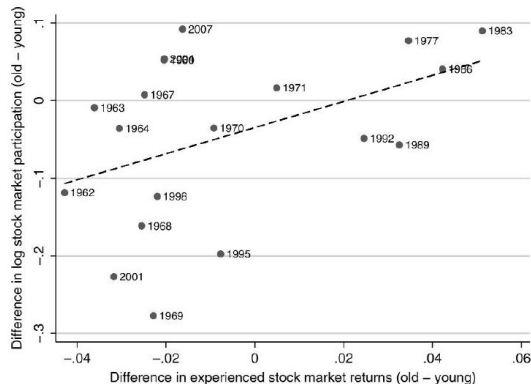


# Sources of Heterogeneous Beliefs

- What may cause heterogeneous beliefs?
- Asymmetric information alone cannot cause rational agents with common prior to agree to disagree
  - e.g., No-trade Theorem of Milgrom and Stockey (1981)
- Heterogeneous beliefs must come from:
  - Differences in prior beliefs
  - Differences in how agents process information (bounded rationality)
  - Overconfidence in private signals

# Sources of Heterogeneous Beliefs — Experience Effects

- One important source of heterogeneous beliefs is personal experience
- When forming beliefs, a rational individual should take account of all available information
- However, mounting evidence suggests that people overweight their personal experience



# Sources of Heterogeneous Beliefs — Experience Effects

- **Inflation expectation** (Malmendier and Nagel, 2016)
- Individuals overweight inflation experienced during their lifetimes when forming expectations about future inflation
- Younger individuals update their beliefs more quickly
- Older individuals place more weight on past inflation experiences

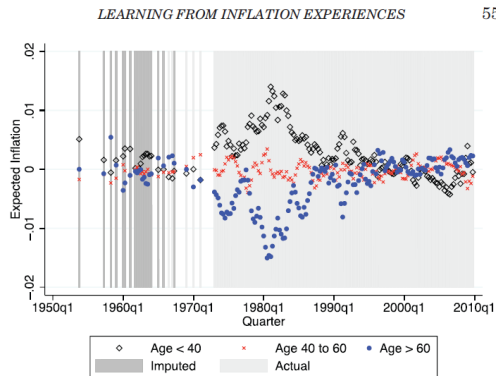
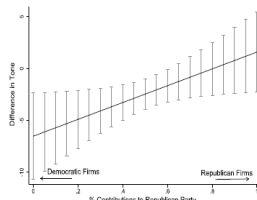
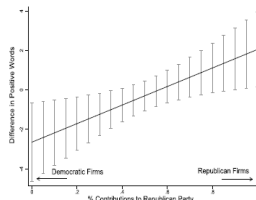


FIGURE I  
Inflation Expectations by Age Group Relative to Cross-Sectional Mean

Malmendier and Nagel (2016)

# Sources of Heterogeneous Beliefs — Political Polarization

- Goldman, Gupta and Israelsen (JFE 2025)
- Setting: Comparing coverage of the same corporate financial news by the conservative Wall Street Journal and the liberal New York Times
- **Findings:**
  - Newspapers use more positive language for firms aligned with their political leanings
  - Effect stronger for politically sensitive topics

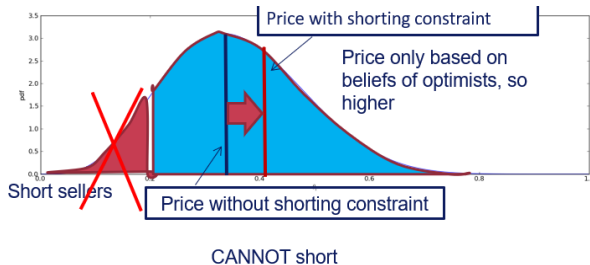


# Heterogeneous Beliefs (no SSC)

- Taken alone, overconfidence-based heterogeneous beliefs do not have interesting pricing effects
- The pricing impacts of optimistic and pessimistic investors cancel out
- But they can help explain the high trading volume
  - Investors trade because they disagree about asset values
  - Higher dispersion of beliefs  $\Rightarrow$  higher trading volume

# Heterogeneous Beliefs with SSC

- If we combine heterogeneous beliefs with short-sale constraints, we get interesting pricing effects
- **Static argument**
- Miller (1977): Heterogeneous Beliefs + Short-Sale Constraint leads to overpricing



# Heterogeneous Beliefs with SSC, ctd.

- **Dynamic argument**
- Harrison and Kreps (1978)
- If there are heterogeneous beliefs, you are willing to pay more than fundamental value today
  - When information is released, there is a good chance that you will be the optimist
  - You can then resell to the other agent at a higher price
  - This creates a “speculative” or “resale option” component to the price

# Heterogeneous Beliefs with SSC, ctd.

- **An example:**  $t = 0, 1, 2$
- Risk-neutral investors A and B trade a risky asset
- Interest rate is zero and no short sale

## Heterogeneous Beliefs with SSC, ctd.

- Both agents' buy-and-hold valuation of the asset is 50
- What is the traded price?
  - In state  $u$ , A sets the price at 90
  - In state  $d$ , B sets the price at 25
  - Then, at time 0, both value the asset at 57.5!
- The price exceeds the expected value because of the resale option

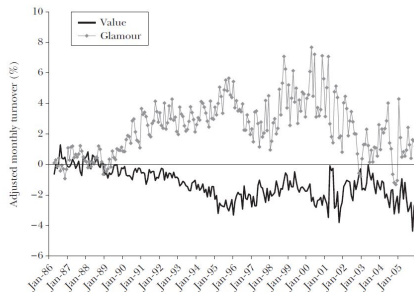
# Heterogeneous Beliefs with SSC, ctd.

- Characteristics of this bubble component:
- The valuation of the resale option is determined by fluctuation of agents' beliefs and asset float
- Scheinkman and Xiong (2003): the more likely their beliefs are to diverge in the future, the larger the bubble
- Float: More shares available for trading  $\Rightarrow$  smaller bubble

# Heterogeneous Beliefs with SSC, ctd.

- Models of heterogeneous beliefs with SSC are popular because they not only explain overpricing, but also another important empirical fact:
- **The coincidence of high valuations and heavy trading**
- Evidence: Hong and Stein (2007)

Figure 2  
Turnover in Value and Glamour Stocks, 1986–2005



# Heterogeneous Beliefs and Bubbles

- Xiong and Yu (AER 2011) conduct a more rigorous test of the heterogeneous beliefs-based bubble theory — The Chinese Warrant Bubble
- In 2005–2008, over a dozen put warrants traded in China went so deep out-of-the-money that their intrinsic value was essentially zero
- Yet they traded at high prices with enormous turnover

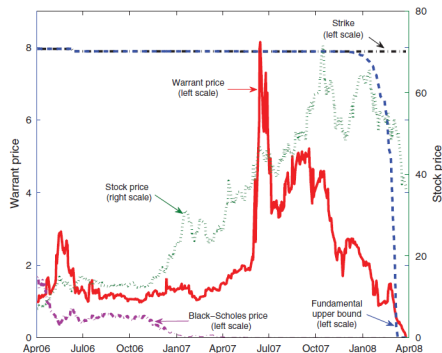


FIGURE 1. PRICES OF WULIANG PUT WARRANT

Notes: This figure shows the daily closing prices of WuLiang stock and its put warrant, along with WuLiang warrant strike price, upper bound of its fundamental value assuming WuLiang stock price drops 10 percent every day before expiration (maximum allowed per day in China's stock market), and its Black-Scholes price using WuLiang stock's previous one-year rolling daily return volatility.

# Heterogeneous Beliefs and Bubbles

- They find evidence most consistent with resale option theory of bubble
- The size of the warrant bubble increases with turnover and volatility, and decrease with asset float

TABLE 5—DETERMINANTS OF SIZE OF THE WARRANTS BUBBLE

|                       | (1)    | (2)    | (3)     | (4)     | (5)     | (6)     |
|-----------------------|--------|--------|---------|---------|---------|---------|
| <i>TURNOVER</i>       | 0.203  |        |         | 0.158   |         | 0.105   |
| ( <i>t</i> -stat)     | (8.91) |        |         | (5.46)  |         | (3.80)  |
| <i>VOL</i>            |        | 21.76  |         | 14.12   |         | 13.99   |
| ( <i>t</i> -stat)     |        | (5.38) |         | (2.97)  |         | (3.05)  |
| <i>FLOAT</i>          |        |        | −0.302  | −0.324  |         |         |
| ( <i>t</i> -stat)     |        |        | (12.62) | (12.31) |         |         |
| <i>FIRMISSUE</i>      |        |        |         |         | −1.302  | −1.260  |
| ( <i>t</i> -stat)     |        |        |         |         | (25.03) | (23.53) |
| <i>BROKERAGEISSUE</i> |        |        |         |         | 0.496   | 0.419   |
| ( <i>t</i> -stat)     |        |        |         |         | (15.84) | (12.60) |

*Notes:* This table includes the regression results of daily warrant closing prices on *TURNOVER* (daily warrant turnover) in column 1, on *VOL* (5min warrant return volatility) in column 2, on *FLOAT* (the total number of warrants outstanding, in billions) in column 3, on *TURNOVER*, *VOL*, and *FLOAT* together in column 4, on *FIRMISSUE* (the total amount of warrants issued by the underlying firm from [Table 1](#) in billions) and *BROKERAGEISSUE* (net total warrant issued by brokerage firms, in billions) in column 5, and on *TURNOVER*, *VOL*, *FIRMISSUE*, and *BROKERAGEISSUE* together in column 6. The regressions use the zero-fundamental sample (defined as Black-Scholes value less than 0.0005, or for cash-settled NanHang if the settlement price will for sure exceed the strike) and include warrant maturity fixed effect. The *t*-statistics, in the parentheses, are adjusted for heteroskedasticity and correlation within a trading day.

Xiong and Yu (AER 2011)

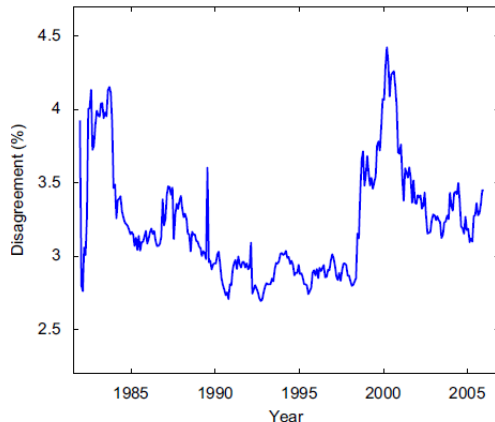
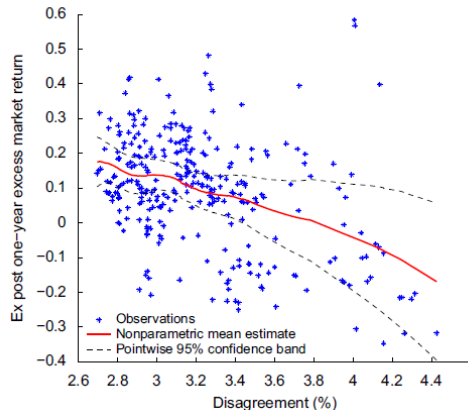
# Heterogeneous Beliefs and Cross-Sectional Stock Returns

- XS prediction of models of heterogeneous beliefs with short-sale constraints:
- Assets that people disagree about more should be more overpriced
- Diether, Malloy, and Scherbina (JF 2003) test the model's prediction using analyst forecast dispersion as a proxy for disagreement
- They find that stocks with higher analyst forecast dispersion earn lower future returns

| Dispersion<br>Quintiles | Mean Returns      |                   |                   |        |        | All<br>Stocks     |
|-------------------------|-------------------|-------------------|-------------------|--------|--------|-------------------|
|                         | Size Quintiles    |                   |                   |        |        |                   |
|                         | Small             |                   | Large             |        |        |                   |
|                         | S1                | S2                | S3                | S4     | S5     |                   |
| D1 (low)                | 1.52              | 1.45              | 1.50              | 1.51   | 1.48   | 1.48              |
| D2                      | 1.12              | 1.40              | 1.41              | 1.18   | 1.35   | 1.36              |
| D3                      | 0.99              | 1.20              | 1.32              | 1.11   | 1.36   | 1.23              |
| D4                      | 0.76              | 1.07              | 1.18              | 1.33   | 1.33   | 1.12              |
| D5 (high)               | 0.14              | 0.56              | 0.83              | 1.03   | 1.20   | 0.69              |
| D1–D5                   | 1.37 <sup>a</sup> | 0.89 <sup>a</sup> | 0.67 <sup>b</sup> | 0.48   | 0.29   | 0.79 <sup>a</sup> |
| <i>t</i> -statistic     | (5.98)            | (3.12)            | (2.41)            | (1.55) | (0.94) | (2.88)            |

# Heterogeneous Beliefs and Aggregate Stock Market

- Yu (2011) tests this prediction for aggregate stock market
- Aggregate disagreement is measured as the value-weighted average of stock-level dispersion



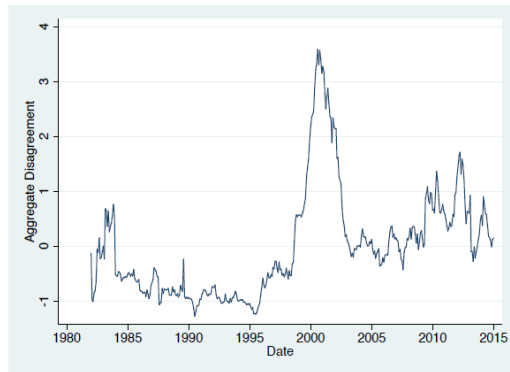
# Heterogeneous Beliefs and Risk-Return Tradeoff

- Another important application: risk-return tradeoff
- Standard theory: higher risk  $\Rightarrow$  higher expected return
- But empirically, the relation between risk and return can be negative
- Heterogeneous beliefs models can explain this anomaly

# Heterogeneous Beliefs and Risk-Return Tradeoff

Figure 1: Time-series of Aggregate Disagreement

Note: Sample Period: 12/1981-12/2014. Sample: CRSP stock file excluding penny stocks (price < \$5) and microcaps (stocks in bottom 2 deciles of the monthly size distribution using NYSE breakpoints). Each month, we calculate for each stock the standard deviation of analyst forecasts on the stock's long run growth of EPS, which is our measure of stock-level disagreement. We also estimate for each stock  $i$   $\hat{\beta}_{i,t-1}$ , the stock market beta of stock  $i$  at the end of the previous month. These betas are estimated with a market model using daily returns over the past calendar year and 5 lags of the market returns. Aggregate Disagreement is the monthly  $\hat{\beta}_{i,t-1}$ -weighted average of stock-level disagreement.



- When investors disagree, the pessimists are sidelined by short-sale constraints
- The optimists set the price
- Higher disagreement  $\Rightarrow$  higher price  $\Rightarrow$  lower expected return

# Heterogeneous Beliefs and Risk-Return Tradeoff

Figure: Average 3-months excess returns for equal-weighted beta decile portfolios in low and high aggregate disagreement months.

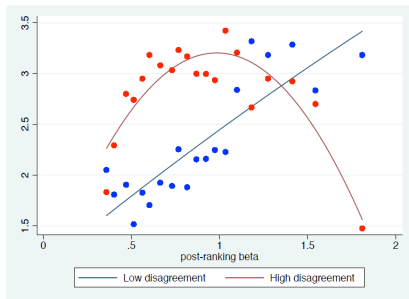
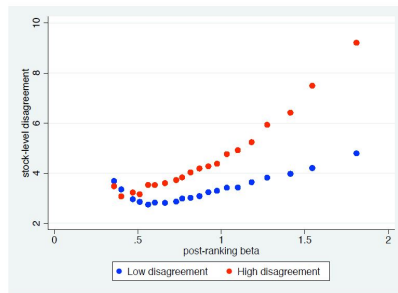
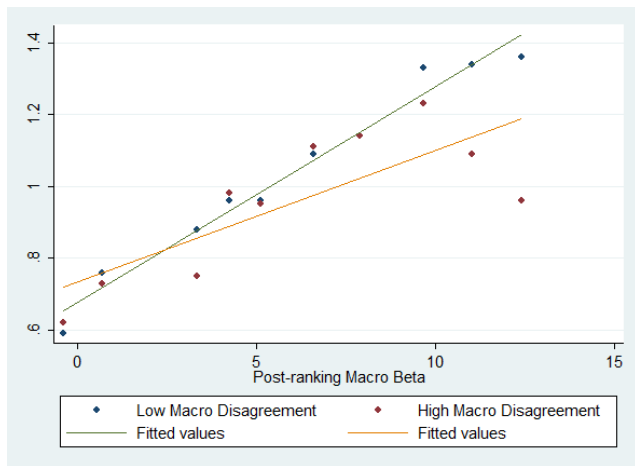


Figure: Equal-weighted average of dispersion of analyst earnings forecasts by beta deciles during low and high aggregate disagreement months.



# Heterogeneous Beliefs and Risk-Return Tradeoff

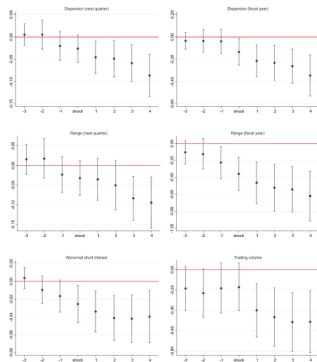
- The same intuition can explain the pricing of macroeconomic factors (Li 2016)



# The Causal Effect of Disagreement on Asset Prices

- Identifying exogenous shocks to investor disagreement is still challenging
- Chang, Hsiao, and Ljungqvist (JF 2022):
  - Exploits the staggered implementation of EDGAR by SEC, which induces a reduction in information acquisition cost
  - This leads to lower disagreement and lower stock prices

# The Causal Effect of Disagreement on Asset Prices



|                            | Skewness<br>(NCSKEW)<br>(1) | Down-to-up<br>volatility<br>(DUVOL)<br>(2) | Extreme<br>negative<br>returns,<br>0.01%<br>(CRASH001)<br>(3) |
|----------------------------|-----------------------------|--------------------------------------------|---------------------------------------------------------------|
| Quarter of EDGAR inclusion | 0.007<br>0.022              | 0.000<br>0.014                             | -0.003<br>0.008                                               |
| Next four quarters         | -0.049**<br>0.020           | -0.031**<br>0.012                          | -0.027***<br>0.007                                            |
| Controls?                  | yes                         | yes                                        | yes                                                           |
| Calendar quarter FE?       | yes                         | yes                                        | yes                                                           |
| Fiscal quarter FE?         | yes                         | yes                                        | yes                                                           |
| Firm FE?                   | yes                         | yes                                        | yes                                                           |
| R-squared                  | 17.4%                       | 19.1%                                      | 15.8%                                                         |
| Pre-trends (p-value)       | 0.562                       | 0.141                                      | 0.957                                                         |
| No. of firms               | 3,366                       | 3,366                                      | 3,366                                                         |
| No. of firm-quarters       | 28,652                      | 28,652                                     | 28,652                                                        |

$$CRASH\ RISK_{i,t} = \alpha + \delta_1 SHOCK_{i,t} + \delta_2 POSTSHOCK_{i,t} + \alpha X_{i,t-1} + c_{i,t} + c_q + c_{i,t} + \xi_{i,t} \quad (2)$$

# Biased Beliefs: Concerns

- Do people simply learn their way out of biased beliefs?
- Only to a limited extent
- Many of the most persistent behavioural biases are useful mental shortcuts for human evolution
  - “Loss aversion” increases survival probability
  - Overconfidence increases effort and social status
- Smart people can be more susceptible to some biases
  - E.g. confirmation bias (Stanovich et al., 2013)

# Stakes and Investor Biases

- Sui and Wang (2025 JFE): investors do better when they care less
- Unique setting where investors trade stocks in real accounts using their own money and simultaneously in a simulated setting
- With lower stakes, investors trade less, diversify more, and earn higher returns

Table 6

Differences between the real and simulated accounts. In this table, we report differences between real and simulated accounts. See [Section 4.1](#) for the bias definitions. For each type of bias we report the mean of the real accounts, the mean of the simulated accounts, and the mean of the differences between them. For the differences, we report *t*-statistics and Wilcoxon *p*-values. In column “N” we report the number of investors.

|                      | N    | Real   | Simulated | Difference |                |                   |
|----------------------|------|--------|-----------|------------|----------------|-------------------|
|                      |      | Mean   | Mean      | Mean       | <i>t</i> -stat | Wilcoxon <i>p</i> |
| Disposition effect   | 4371 | 4.163  | 0.409     | 3.674      | (32.43)        | 0.000             |
| Lottery preference   | 4369 | 6.491  | 4.264     | 2.228      | (13.00)        | 0.000             |
| Extrapolation        | 4373 | 15.599 | 12.66     | 2.939      | (6.49)         | 0.006             |
| Underdiversification | 4394 | −1.006 | −1.473    | 0.386      | (32.67)        | 0.000             |
| Local Bias           | 1739 | 0.630  | 0.645     | −0.015     | (−0.64)        | 0.010             |
| Turnover             | 4306 | 5.960  | 0.839     | 5.121      | (47.44)        | 0.000             |
| Performance          | 4375 | −0.656 | −0.140    | −0.509     | (−4.95)        | 0.001             |